

# Problems on Celestial Mechanics

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## Binary System

*The aim of this problem is to build a model that describes the dynamics of a binary system composed of two celestial bodies that exchange matter.*

Consider a binary system formed by two celestial bodies with masses  $m_1$  and  $m_2$ . For the first part of the problem (tasks 1 to 4), assume that  $m_1 \gg m_2$ .

1. Find the angular velocity  $\omega$ , the angular momentum  $\vec{L}$ , and the mechanical energy  $E$  of the system, in the case where the lighter body moves in a circular orbit of radius  $R$ .
2. Find the angular momentum  $\vec{L}$  and mechanical energy  $E$  of the system, in the case where the lighter body moves in an elliptical orbit with semi-major axis  $a$  and eccentricity  $\epsilon$ .

The system is immersed in a region of space filled with interstellar dust, stationary in the coordinate system centered on the heavier body.

3. During its motion, the mass of the lighter body increases due to inelastic collisions with interstellar dust. Which quantities of the system are conserved during the orbital motion?
4. Starting from the energy and force balance, it can be shown that the rate of change of mechanical energy is given by

$$\frac{dE}{dt} = - \left[ \frac{E}{m_2} + \frac{2Gm_1}{r(t)} \right] \frac{dm_2}{dt},$$

where  $r(t)$  is the instantaneous distance between the two bodies and  $\frac{dm_2}{dt}$  is the rate of change of the mass of the lighter body. Assuming  $m_2$  varies very slowly, show that the eccentricity of the orbit, defined as

$$\epsilon = \sqrt{1 + \frac{2EL^2}{G^2m_1^2m_2^3}},$$

remains constant during the process. *Hint:* For an elliptical orbit, it holds that  $\frac{1}{T} \int_0^T \frac{dt}{r(t)} = \frac{1}{a}$ , where  $T$  is the orbital period and  $a$  is the semi-major axis.

5. How do the results of the previous points change if the assumption  $m_1 \gg m_2$  no longer holds?

Now suppose that both bodies, on circular orbits, exchange matter: due to an unclear transfer process, at every instant one of the two bodies loses a certain amount of mass, which is captured by the other. This process is caused only by internal forces within the system.

6. Show that

$$\frac{1}{2R} \frac{dR}{dt} = \frac{dm_1}{dt} \left[ \frac{m_1(t) - m_2(t)}{m_1(t) m_2(t)} \right],$$

where  $R(t)$  is the sum of the radii of the two circular orbits.

7. Show that the mechanical energy of the system decreases during the process if the more massive body is the one losing mass.
8. Now let  $m_1(t) = m_0 + kt$  and  $m_2(t) = m_0 - kt$ . What is the distance between the two bodies at a generic time  $t < m_0/k$ ?
9. Referring to the situation described in the previous task, what is the final energy of the system for  $t > m_0/k$ ?

## Ballistic Missile

A ballistic missile is launched from a point  $A$  on the Earth's surface with an initial velocity  $\vec{v}_0$ , whose magnitude is below the escape velocity, and at an angle  $\alpha$  with respect to the horizontal. After following an elliptical trajectory, the missile crashes at a point  $B$ , whose position depends on the launch conditions. The situation is illustrated in Fig. 1.

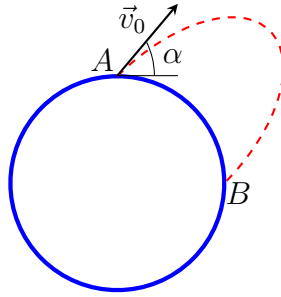


Figure 1: Example of missile motion in the Earth's gravitational field.

1. Find the parameters of the elliptical orbit as functions of  $v_0$  and  $\alpha$ .
2. Determine the flight time and the distance between the launch and impact points on the Earth's surface. How far is the highest point of the trajectory from the Earth's surface?
3. What should the initial velocity  $v_0$  be for a missile launched at a given angle  $\alpha$  to reach the antipode, as shown in Fig. 2?

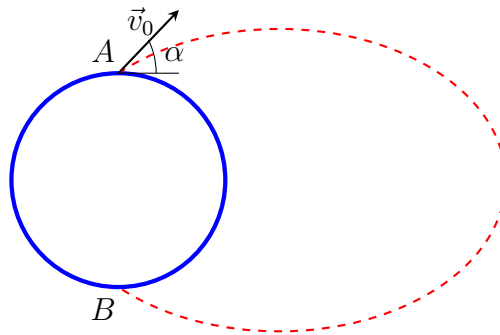


Figure 2: Missile trajectory towards the antipode.